Answer Key to the Theoretical Tests

1. Part A

No.	A	B	C	D	E	F	No.	A	B	C	D	E	F	No.	A	B	C	D	E	F
A1					X		A21		X					A41	X					
A2		X					A22			X				A42						X
A3			X				A23	X						A43			X			
A4					X		A24			X				A44		X				
A5				X			A25				X			A45		X				
A6					X		A26		X					A46		X				
A7				X			A27	X						A47						X
A8				X			A28			X				A48		X				
A9					X		A29		X					A49					X	
A10			X				A30				X			A50			X			
A11			X				A31			X				A51				X		
A12			X				A32			X				A52				X		
A13			X				A33				X			A53		X				
A14			X				A34			X				A54						X
A15			X				A35						X							
A16			X				A36				X									
A17				X			A37			X										
A18		X					A38			X										
A19			X				A39			X										
A20					X		A40				X									

2. Part B

B1. (0.5 points x 6 = 3 points)

	A	B	C	D	E	F
Plants		**X**		**X**		
Mammals	**X**		**X**		**X**	**X**

B2. (0.5 points x 5 = 2.5 points)

I	II	III	IV	V
D	**C**	**E**	**B**	**A**

B3. (0.5 points x 6 = 3 points)

$$\frac{735}{3}\times 122-\left\{\left(\frac{735}{3}-1\right)\times 18\right\}-\{5\times(2\times 1)\}$$
$$=245\times 122-244\times 18-5\times 2$$
$$=29890-4392-10$$
$$=25488$$

When the N-terminal Met residue is assumed to have been removed,

$$\left(\frac{735}{3}-1\right)\times 122-\left\{\left(\frac{735}{3}-1-1\right)\times 18\right\}-\{5\times(2\times 1)\}$$
$$=244\times 122-243\times 18-5\times 2$$
$$=29768-4374-10$$
$$=25384$$

Partial point:
(0.5) Division by 3.
(0.5) Subtraction of water mass of peptide bonds.
(0.5) Number of peptide bonds: number of amino acids *minus* 1.
(0.5) Correct molecular mass of one water molecule.
(0.5) Subtraction of ten hydrogen molecules of five disulfide bonds.
(0.5) Correct calculations.

B4. (3.5 points)

(1) (0.3 points x 10 = 3 points)

A	**6**
B	**1, 3, 7, 10**
C	
D	**4, 9**
E	**2, 5, 8**
F	

(2) (0.5 point : if all are correct)

1	2	3
B	**C**	**A**

B5. (0.5 points x 4 = 2 points)

I	II	III	IV
D	**A**	**C**	**B**

B6. (3 points)

2000

(or 2007 by using more precise Avogadro's number.)

B7. (1 points x 4 = 4 points)

I	II	III	IV
A	**F**	**E**	**A**

B8. (3 points)

	Deficiency symptoms (0.3x3)	Mineral mobility (0.3x3)	Metabolic roles (0.4x3)
Mg	**B**	**C**	**G**
Fe	**A**	**D**	**E**
N	**B**	**C**	**F**

B9. (1 points x 3 = 3 points)

I	II	III
C	**D**	**F**

B10. (4 points)

(1) (0.5 points x 4 = 2 points)

		can respond to Florigen.	cannot respond to Florigen.
The shoot apical meristems of	cold-treated annual plants	**X**	
	untreated annual plants	**X**	
	cold-treated biennial plants	**X**	
	untreated biennial plants	**X**	

(2) (0.5 points x 4 = 2 points)

		produce Florigen under the long-day condition.	do not produce Florigen in either photoperiodic condition.
The leaves of	cold-treated annual plants	**X**	
	untreated annual plants	**X**	
	cold-treated biennial plants	**X**	
	untreated biennial plants		**X**

B11. (3 points)

(1) (1 point)

A	B	C	D
		X	

(2) (1 point)

A	B	C	D
		X	

(3) (1 point)

A	B	C	D
			X

B12. (1 points x 3 = 3 points)

	A	B	C
I	**X**		
II		**X**	
III	**X**		

B13. (1 points x 3 = 3 points)

I	II	III
A	**C**	**B**

B14. (2.5 points)

(1) (0.5 point)

A	B
	X

(2) (0.5 point)

A	B	C	D

		X	

(3) (0.5 points x 3 = 1.5 points)

C	**E**	**G**

B15. (1 points x 4 = 4 points)

	Hypothesis 1	Hypothesis 2	Hypothesis 3	Hypothesis 4
Rejected	**X**	**X**		
Not rejected			**X**	**X**

B16. (2 points)

(1) (1 point)

A	B	C	D	E
	X			

(2) (1 point)

A	B	C	D	E
X				

B17. (2 points)

(1) (0.2 point x 4 = 0.8 points)

	A	B	C	D
TRUE	**X**	**X**		**X**
FALSE			**X**	

(2) (0.3 points x 4 = 1.2 points)

	A	B	C	D
TRUE	**X**		**X**	**X**
FALSE		**X**		

B18. (3 points)

(1) (1 points x 2 = 2 points)

A	B	C	D	E
X			**X**	

(Choosing more than 2 gives no point.)

(2) (1 point)

A	B	C	D
			X

B19. (3 points)

(1) (0.5 point)

A	B	C	D	E

			X	

(2) (0.5 point)

A	B	C	D
			X

(2) (1 points x 2 = 2 points)

A	B	C	D	E	F
		X		**X**	

(Choosing more than 2 gives no point.)

B20. (3 points)

(1) (1 points x 2 = 2 points)

Acc	**190** m
Aml	**340** m

(2) (1 point)

A	B	C	D
	X		

B21. (0.5 points x 4 = 2 points)

i	ii	iii	Iv
C	**D**	**B**	**A**

B22. (2 points)

(1) (1 point)

A	B	C	D
	X		

(2) (1 point)

A	B	C	D	E	F
X					

B23. (4 points)

(1) (2 points)

A	B	C	D	E	F
	X				

(2) (2 points)

A	B	C	D	E
		X		

B24. (4 points)

(1) (1 point)

A	B	C	D	E
	X			

(2) (1 point)

A	B	C	D	E
		X		

(3) (2 points)

26	%

B25. (3 points)
(1) (2 points)

2.5	**X**	**10^6**
(**2,500,000**)		years

5 X 10^6 (**5,000,000**) years: 1 point

(2) (1 point)

A	B	C
		X

B26. (3 points)
(1) (1 point)

A	B	C	D
			X

(2) (1 point)

A	B	C	D
			X

(3) (1 point)

A	B	C	D
		X	

B27. (4 points)
(1) (2 points)

34	%

or 0.34

(2) (1 point)

I	II	III	IV
			X

(3) (1 point)

A	B	C	D

X			

B28. (1 points x 3 = 3 points)

A	B	C	D	E	F	G	H
	X			**X**	**X**		

B29. (3 points)

(1) (0.5 point x 2 = 1 points)

A	B	C	D	E	F	G
		X	**X**			

(Choosing more than 2 gives no point.)

(2) (1 point)

A	B	C	D	E	F	G
	X					

(3) (1 point)

A	B	C	D	E	F	G
X						

B30. (1 points x 3 = 3 points)

I	II	III
D	**C**	**B**

B31. (0.5 points x 5 = 2.5 points)

	A	B	C	D	E
TRUE	**X**	**X**		**X**	
FALSE			**X**		**X**

B32. (1 points x 3 = 3 points)

	A	B	C
TRUE			**X**
FALSE	**X**	**X**	

B33. (0.5 points x 6 = 3 points)

I	II	III	IV	V	VI
D	**C**	**B**	**D**	**E**	**B**

B34. (5 points)

(1) (1 point)

A	B	C	D	E	F	G	H
		X					

(2) (0.5 points x 6 = 3 points)

	Domain I	Domain II	Function
Older	**3**	**1**	**4**
Newer	**1**	**4**	**1**

(3) (1 point: if all are correct)

I	II	III
B	**C**	**A**

B35. (4 points)

(1) (2 points)

A	B	C
	X	

(2) (1 point)

2, 6

Since the number of correct choice is not stated in the question, the points are calculated by (number of the correct answer) x 0.5 – (number of the wrong answer) x 0.3.

(3) (1 point)

A	B	C	D	E	F	G	H	I
				X				

Since the answer of (3) is consequence of the question (1), the combination of the wrong answer (1) A and the answer of (3) B gives 0.5 point for (3), and that of (1) C and (3) H, gives 0.5 point for (3).